



Tracking the tightness: Sep 2026

Distributor channel checks: Pricing and lead times still trending positively

In partnership with dbDataInsights, we have been conducting a weekly scrape of chip pricing of mainly analog/discrete/MCU components sold into the distributor channel (>1,500 SKUs). During September, the trend remains positive - mature chip pricing trending upwards as shortage conditions prevail in specific, lagging-edge categories, albeit overall tightness is not pervasive as in 2021/22. Price increases continue to be significant in parts of analog (via TI/ADI), while it is notable that many mature foundries are raising pricing into 2027, price increases that are likely to be passed on. As per Fusion, Infineon supply constraints are also set to persist into 2027, while core MCU and sensor categories at ST Micro are seeing extended lead times (to 40-50 weeks). Passives (eg MLCCs) have entered shortage conditions.

Leading-edge tightness: Acute shortage continues despite memory de-speccing

The leading-edge logic and memory shortage remains acute. Lead times on N2/N3 are estimated by some observers to range between 78-156 weeks (1 ½ - 3 years), while we believe foundry supply remains insufficient to cover demand at least until the end of 2028. On the memory side which was arguably even more constrained until recently, we see the recent de-speccing moves as a rational response to a heavily constrained memory market (effectively driving more affordable xPU volumes at the expense of driving HBM to 12/16-hi). However, relief from the shortage does not look imminent. For example, Fusion notes that fulfilment rates on DDR5 DRIMM modules for non-hyperscaler customers sits today below 30%, while legacy DDR3/DDR4 still faces EOL challenges. Server CPUs also remain particularly tight with ABF substrate lead times stretched at over a year.

Tracking the AI trade: Introducing our AI trade monitor

In light of industry nervousness around the AI trade which is tangential, but still critical to sentiment in semis investing, we have added some data to this month's edition of "*Semis Skimmed*" to address common questions around AI model performance and monetization, token pricing, GPU rental pricing, the rise of open weight, along with broader financial risks. We also have added a section on industrial constraints downstream. We remain constructive on AI-related names and believe that the existential risk facing the hyperscalers is too great to see hyperscaler capex flatlining in 2028, which is the principal risk factor facing semis investing at present. However, as we update the data set in the coming months, we will look for any major deviations in the data that could upset that assumption.

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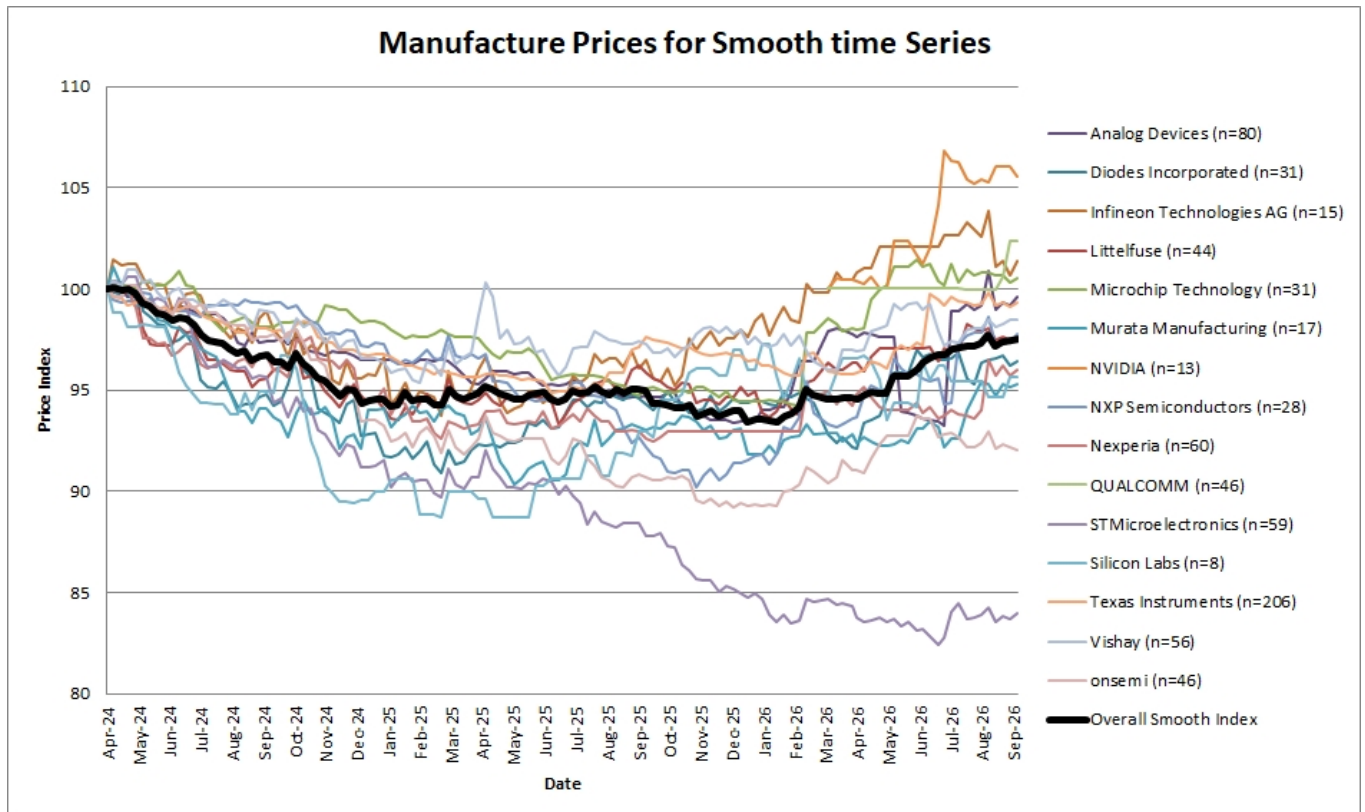
1. Disti checks

Figure 1: Chip pricing (indexed to 100, by vendor, end of period)

	24Q1	24Q2	24Q3	24Q4	25Q1	25Q2	25Q3	25Q4	26Q1	26Q2	26Q3
Allegro Microsystems (n=6)	100.0	98.8	93.1	92.1	94.9	95.7	93.9	92.0	90.6	90.6	92.1
Altera (n=32)				100.0	102.9	102.9	101.2	101.2	101.0	101.0	101.0
Analog Devices (n=80)	100.0	99.2	97.5	96.5	95.3	95.3	94.7	93.5	98.0	93.2	99.6
Diodes Incorporated (n=31)	100.0	97.5	94.4	92.9	92.0	93.9	94.4	94.5	92.7	96.5	96.5
Infineon Technologies AG (n=15)	100.0	99.1	97.5	95.7	95.1	94.6	95.5	98.4	100.4	102.6	101.4
Intel (n=13)					100.1	104.5	109.6	109.6	107.2	108.1	108.5
Littelfuse (n=44)	100.0	98.1	96.1	94.6	94.3	93.8	95.4	94.9	96.0	97.1	97.6
Microchip Technology (n=31)	100.0	100.9	98.3	98.4	97.6	95.7	95.0	94.5	97.9	100.2	100.6
Murata Manufacturing (n=17)	100.0	98.1	93.4	94.0	92.8	90.9	93.1	91.8	92.5	92.2	95.3
NVIDIA (n=13)									100.4	106.8	105.6
NXP Semiconductors (n=28)	100.0	98.9	99.3	97.3	96.8	94.8	91.6	91.8	93.3	94.3	97.8
Nexperia (n=60)	100.0	96.9	95.9	94.5	93.3	93.5	92.7	93.0	94.7	93.7	96.0
QUALCOMM (n=46)									100.0	100.0	102.4
STMicroelectronics (n=59)	100.0	99.5	94.6	91.3	90.7	90.2	88.0	84.9	84.5	82.8	83.9
Silicon Labs (n=8)	100.0	95.9	96.7	90.0	90.0	90.6	94.5	96.0	96.6	96.2	95.7
Texas Instruments (n=206)	100.0	99.0	98.0	96.8	95.7	94.9	97.4	96.5	95.8	99.2	99.3
Vishay (n=56)	100.0	100.0	98.2	96.6	96.9	96.1	96.9	97.6	97.0	97.2	98.5
onsemi (n=46)	100.0	99.4	97.4	93.5	92.2	91.7	90.6	89.4	91.5	92.8	92.0
Overall Smooth Index	100.0	98.6	96.4	94.5	94.7	94.6	94.3	93.5	94.6	96.8	97.5

Source : dbDataInsights, Deutsche Bank; data to September 9th 2026; note that outliers are smoothed; number in brackets indicates number of products scraped in the latest scrape

Figure 2: Chip pricing (indexed to 100), by vendor, weekly



Source : dbDataInsights, Deutsche Bank; data to September 9th 2026; note that outliers are smoothed; number in brackets indicates number of products scraped in the latest scrape



Pricing trend by mature chip category

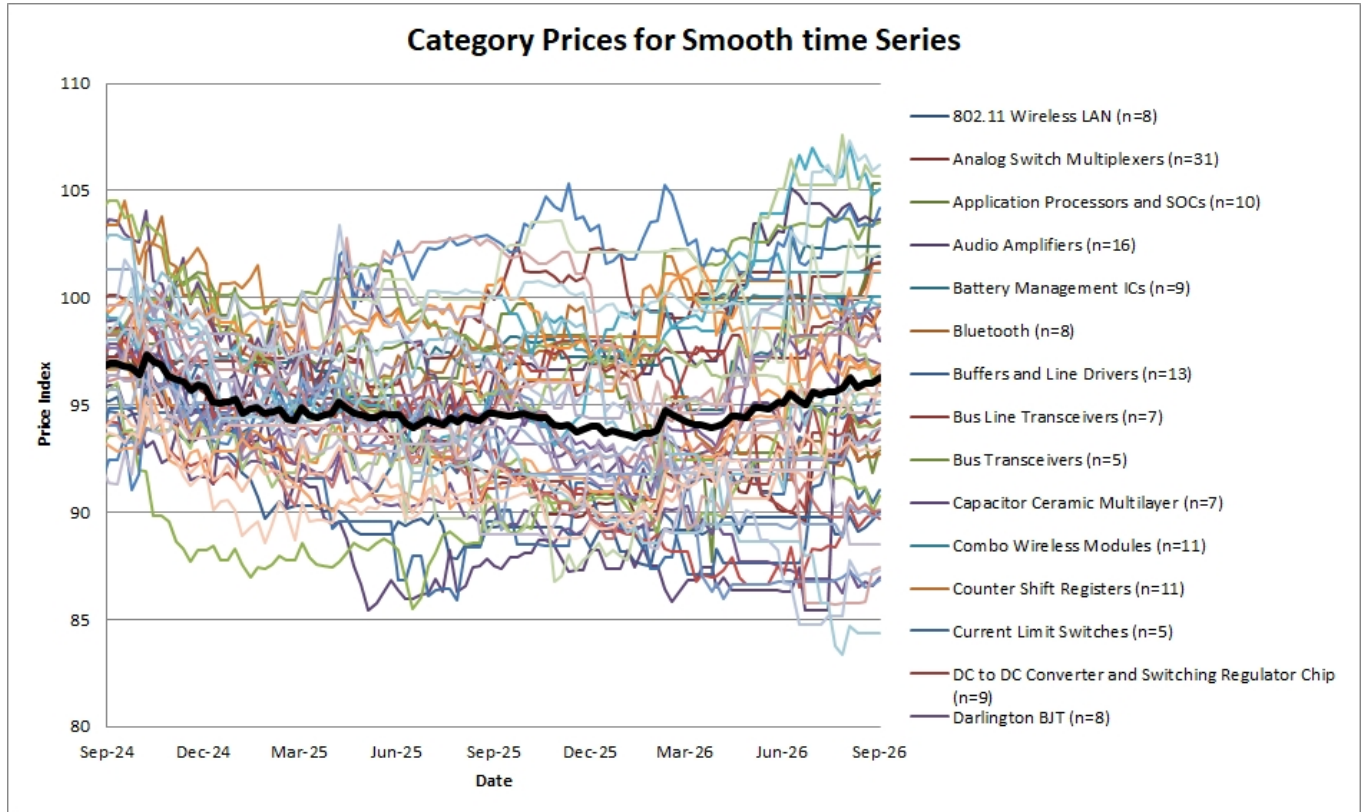
Figure 3: Chip pricing index (to 100), by product category, end of period

	24Q1	24Q2	24Q3	24Q4	25Q1	25Q2	25Q3	25Q4	26Q1	26Q2	26Q3
802.11 Wireless LAN (n=8)									100.0	100.0	101.9
Analog Switch Multiplexers (n=31)	100.0	98.1	96.7	95.2	94.7	93.1	91.7	90.4	91.8	89.5	92.9
Application Processors and SOC's (n=10)									100.0	100.0	105.3
Audio Amplifiers (n=16)	100.0	99.0	97.7	96.1	96.7	92.4	96.9	97.2	97.0	104.4	103.6
Battery Management ICs (n=9)		99.3	96.7	97.0	96.6	94.4	98.2	97.3	94.8	102.5	102.4
Bluetooth (n=8)									100.0	100.0	101.6
Buffers and Line Drivers (n=13)	100.0	96.5	94.6	92.8	90.0	89.7	89.3	89.1	89.0	89.8	90.1
Bus Line Transceivers (n=7)	100.0	101.0	97.3	94.5	95.6	97.9	102.6	102.3	100.2	98.4	101.7
Bus Transceivers (n=5)	100.0	98.3	96.9	99.2	97.1	96.7	99.7	96.3	92.8	92.7	93.1
Capacitor Ceramic Multilayer (n=7)	100.0	98.9	94.7	94.1	89.8	86.1	87.9	87.3	87.4	85.4	98.0
Combo Wireless Modules (n=11)									100.0	100.0	100.0
Counter Shift Registers (n=11)	100.0	99.1	97.3	98.6	96.1	96.0	97.8	96.9	93.4	93.3	92.7
Current Limit Switches (n=5)		98.1	93.5	93.8	96.3	94.3	91.9	89.8	87.1	88.4	91.1
DC to DC Converter and Switching Regulator Chip (n=9)	100.0	103.1	99.7	97.2	96.9	94.5	95.5	96.8	97.7	97.2	99.4
Darlington BJT (n=8)	100.0	106.4	102.8	98.7	97.4	98.0	95.1	92.1	92.8	94.6	95.6
Decoders and Demuxes (n=5)	100.0	101.5	97.7	95.4	100.1	102.6	102.0	102.0	91.7	92.3	95.6
Digital BJT - Pre-Biased (n=8)	100.0	98.2	103.0	100.7	99.8	99.1	96.1	98.2	100.8	95.5	96.9
Digital Isolators (n=14)	100.0	101.3	94.9	94.6	91.6	88.0	88.5	90.8	91.7	93.3	94.6
Digital to Analog Converters - DACs (n=7)	100.0	100.8	98.6	97.1	95.3	94.9	92.0	90.9	97.5	90.1	93.9
EEPROM (n=14)	100.0	100.3	99.3	100.1	100.9	99.5	97.6	97.1	101.5	103.5	103.5
ESD Suppressors (n=16)	100.0	99.4	98.6	92.6	91.7	90.3	89.4	89.4	89.7	86.9	87.0
Ethernet Controllers (n=6)											100.0
Field Programmable Gate Arrays - FPGAs (n=31)				100.0	102.9	102.9	102.9	102.9	102.5	102.5	102.5
Flash (n=0)				98.9	97.1	97.3	97.6	102.6	102.9	105.9	105.9
Flip Flops (n=12)	100.0	94.3	93.7	91.8	94.1	95.4	97.3	98.0	92.3	93.2	94.2
Fuses (n=14)	100.0	96.6	93.8	87.8	87.5	86.0	89.1	91.0	92.5	92.5	94.3
GP BJT (n=22)	100.0	98.3	97.2	94.6	91.9	93.5	93.7	93.0	93.6	97.6	96.9
GPIO Expanders (n=1)							99.2	98.5	98.5	101.2	101.2
Gate and Power Drivers (n=15)	100.0	100.8	96.3	93.7	93.6	92.6	97.0	95.6	99.4	98.2	99.7
Inductor Surface Mount (n=6)	100.0	95.5	91.0	93.2	95.4	101.7	102.2	101.7	101.2	101.5	104.2
Inverters and Schmitt Triggers (n=11)	100.0	97.6	97.4	94.8	93.9	95.0	91.5	89.1	87.1	87.4	89.7
LED Drivers (n=10)	100.0	100.2	103.9	98.4	95.1	93.1	92.5	90.7	90.0	94.9	90.8
Latches (n=7)		92.6	88.3	87.7	89.3	87.3	87.7	83.9	85.7	83.5	86.0
Level Translators (n=14)	100.0	99.6	98.0	97.2	94.9	94.7	95.0	97.6	98.9	106.0	105.0
Linear Comparators (n=12)	100.0	101.4	99.7	98.9	97.8	99.1	100.0	97.7	101.4	95.2	95.9
Linear Regulators (n=10)	100.0	98.0	93.3	94.5	95.1	94.6	95.6	91.8	87.0	86.8	86.8
Logic Gates (n=13)	100.0	98.9	99.2	94.9	91.8	92.4	91.5	88.8	93.2	91.3	90.1
MOSFETs (n=14)	100.0	97.0	93.7	93.7	92.8	93.3	95.0	98.1	97.1	98.1	96.5
Microcontrollers - MCUs (n=10)	100.0	99.6	96.7	94.6	95.3	94.5	93.2	91.2	94.6	98.2	99.6
Microprocessors - MPUs (n=16)									99.7	99.7	99.6
Motor Controller and Driver ICs (n=12)	100.0	98.3	93.0	92.8	90.4	90.5	91.2	91.0	92.6	96.2	96.9
Multiplexers (n=6)	100.0	99.4	101.3	94.2	96.1	91.5	91.8	91.8	89.4	89.4	89.9
Operational Amplifiers - Op Amps (n=16)	100.0	98.0	96.2	96.6	94.7	93.0	95.7	92.6	95.4	94.9	98.2
PHY (n=6)									99.8	105.3	105.6
PTC Resettable Fuses (n=8)	100.0	98.7	97.8	96.2	96.0	95.9	95.7	93.2	92.2	92.4	93.4
Power Switches (n=5)		100.1	102.7	95.2	95.3	97.5	97.5	94.2	92.2	94.0	95.2
RF Transceivers (n=6)											101.3
Rectifiers (n=17)	100.0	99.3	94.9	96.2	94.1	95.1	94.0	92.7	92.1	92.6	93.9
Resistor Fixed Single-Through Hole (n=10)		103.0	99.6	97.6	97.9	102.6	101.9	97.8	90.8	85.8	87.4
Signal Buffers and Repeaters (n=5)					98.0	92.2	89.0	88.2	95.0	95.1	95.5
Special Purpose Amplifiers (n=10)	100.0	97.7	96.1	98.3	97.9	99.8	97.6	94.4	95.2	98.6	98.6
Supervisory Circuits (n=7)	100.0	100.1	98.5	94.4	94.5	91.8	92.7	89.7	89.1	88.0	84.4
Transient Voltage Suppressors - TVS Diodes (n=20)	100.0	99.0	94.0	92.2	91.7	90.6	91.3	89.9	91.9	91.5	93.0
Transistor and Photovoltaic Output Photocouplers (n=5)			101.3	99.1	99.9	97.4	96.2	95.1	92.0	84.7	87.3
USB Interface ICs (n=4)	100.0	95.9	98.9	94.1	94.1	93.9	95.9	96.0	95.0	95.7	96.7
USB Power Switches (n=4)						99.6	102.4	102.1	101.0	98.8	102.1
Varistors (n=4)	100.0	100.1	91.7	96.1	92.9	93.2	89.0	91.6	91.6	92.0	88.5
Voltage References (n=13)	100.0	99.3	100.5	97.5	97.2	99.0	100.0	99.7	99.5	102.5	106.2
Zener (n=19)	100.0	96.7	93.7	90.6	89.6	90.0	90.8	89.3	92.3	93.6	95.5
Overall Smooth Index	100.0	98.8	96.8	95.1	94.5	94.2	94.6	93.8	94.1	95.0	96.3

Source : dbDataInsights, Deutsche Bank; data to September 9th 2026; note that outliers are smoothed; number in brackets indicates number of products scraped in the latest scrape



Figure 4: Chip pricing (indexed to 100), by product category, weekly



Source : dbDataInsights, Deutsche Bank; data to September 9th 2026; note that outliers are smoothed; number in brackets indicates number of products scraped in the latest scrape



Availability trend by mature chip category

Figure 5: Mature chip lead time monitor

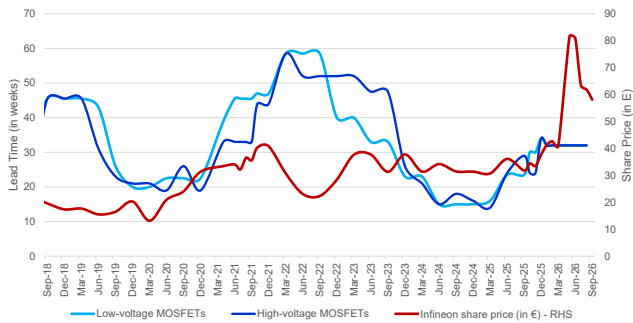
Vendor	Product	Go-forward pricing trend				Lead times (weeks)								Go-forward lead-time trend			
		Late-May	Mid-Jun	Mid-Jul	Mid-Aug	2018 peak	2022 Peak	Early-Mar	Late-May	Mid-Jun	Mid-Jul	Mid-Aug	Late-May	Mid-Jun	Mid-Jul	Mid-Aug	
Infineon	Sensors	↗	↗	↗	↗	16-39	18-52	4-26	4-60	4-60	4-60	4-60	↗	↗	↗	↗	
	Switching regulators	↗	↗	↗	↗	30+	40-52	16-24	16-24	16-24	16-24	16-24	↗	↗	↗	↗	
	Bluetooth modules	→	↗	↗	↗	NA	26-36	16-24	16-24	16-24	16-24	16-24	→	↗	→	→	
	Low-voltage MOSFETs	↗	↗	↗	↗	26-38	52-65	12-52	12-52	12-52+	12-52+	12-52+	↗	↗	↗	↗	
	High-voltage MOSFETs	↗	↗	↗	↗	22-26	50-54	12-52	12-52	12-52+	12-52+	12-52+	↗	↗	↗	↗	
	IGBTs	↗	↗	↗	↗	26-39	39-50	10-44	12-52	12-52	12-52	12-52	↗	↗	↗	↗	
	8-bit MCU	→	↗	↗	↗	16-18	45-52	10-26	10-26	10-26	10-26	10-26	→	→	→	→	
	32-bit MCU	→	↗	↗	↗	20-24	45	10-97	10-97	10-97	10-97	10-97	↗	↗	↗	↗	
	Automotive high-end NOR flash	→	→	→	→	20-32	32-45	32-45	32-45	32-45	32-45	32-45	→	→	→	→	
Melexis	Sensors	↘	↘	↘	↘	Allocation	32-52	12-26	12-26	20-26	20-26	20-26	↗	↗	↗	↗	
Murata	WiFi modules	→	→	→	→	22-26	52-72	26-50	26-50	26-50	26-50	26-50	↗	↗	↗	↗	
	Bluetooth modules	→	→	→	↗	22-26	52-64	26-50	26-50	26-50	26-50	26-50	↗	↗	↗	↗	
NXP	Interface	↗	↗	↗	↗	10-12	36-52	16-20	16-20	16-20	16-20	16-20	→	→	→	→	
	8-bit MCU	SMA	→	↗	↗	14-16	Allocation	16-39	16-39	16-39	16-39	16-39	→	→	→	→	
	32-bit MCU	SMA	→	↗	↗	20-24	Allocation	16-39	16-39	16-39	16-39	16-39	→	→	→	→	
	32-bit MPU	SMA	→	↗	↗	16	Allocation	18-39	18-39	18-52	18-52	18-52	→	→	↗	↗	
	Automotive high-end	SMA	→	↗	↗	12-18	Allocation	18-52	18-52	18-52	18-52	18-52	→	→	→	→	
onsemi	Sensors	↗	↗	→	→	18-24	18-52	18-52	18-52	18-52	18-52	18-52	↗	↗	→	→	
	Amplifiers and converters	↗	↗	↗	↗	12-24+	35-42	10-16	10-16	10-16	10-16	10-16	→	→	→	→	
	Timing	↗	↗	↗	↗	10-12	35-42	22-32	22-32	22-32	22-32	22-32	→	→	→	→	
	Multi-source analog/power	↗	↗	↗	↗	30+	35-42	12-22	12-22	12-22	12-22	12-22	↗	↗	↗	↗	
	Switching regulators	↗	↗	↗	↗	15+	35-42	14-26	14-26	14-26	14-26	14-26	↗	↗	↗	↗	
	Bluetooth modules	→	→	→	→	NA	16-30	16-30	16-30	16-30	16-30	16-30	→	→	→	→	
	Low-voltage MOSFETs	↗	↗	↗	↗	30-42	47-52	10-52	10-52	10-52	10-52	10-52	↗	↗	↗	↗	
	High-voltage MOSFETs	↗	↗	↗	↗	16-26	36-52	7-50	10-50	10-52	10-52	10-52	↗	↗	↗	↗	
	IGBTs	↗	↗	↗	↗	20-52	39-52	8-31	12-41	8-32	8-32	8-32	↗	↗	↗	↗	
Renesas	Amplifiers and converters	↗	↗	↗	↗	8-10	50-60	14-16	14-16	14-16	14-16	14-16	↗	↗	↗	↗	
	Timing	↗	↗	↗	↗	8-10	50	20-26	20-26	20-26	20-26	20-26	↗	↗	↗	↗	
	Interface	↗	↗	↗	↗	14-16	40-50	12-14	12-14	12-14	12-14	12-14	↗	↗	↗	↗	
	Switching regulators	↗	↗	↗	↗	NA	40-50	18-26	18-26	18-36	18-36	18-36	↗	↗	↗	↗	
	8-bit MCU	→	→	→	→	25	52	14-18	18	18	18	18	↗	↗	↗	↗	
	32-bit MCU	→	→	→	→	24-26	52	14-18	18	18	18	18	↗	↗	↗	↗	
	Automotive high-end	→	→	→	→	30	45	24	24	24	24	24	→	→	→	→	
ST Micro	Sensors	↗	↗	↗	↗	14-20	36-40	20-34	20-34	24-52	24-52	24-52	↗	↗	↗	↗	
	Amplifiers and converters	↗	↗	↗	↗	24+	45-52	20-22	20-22	20-22	20-22	28-30	↗	↗	↗	↗	
	Multi-source analog/power	↗	↗	↗	↗	30+	40-50	20-24	20-24	20-24	20-24	30-36	↗	↗	↗	↗	
	Switching regulators	↗	↗	↗	↗	25+	40-50	18-26	18-26	18-26	18-26	30-52+	↗	↗	↗	↗	
	Low-voltage MOSFETs	→	↗	↗	↗	38-42	50-54	10-23	10-23	10-23	10-23	10-23	↗	↗	↗	↗	
	High-voltage MOSFETs	→	↗	↗	↗	38-42	47-52	13-26	12-26	12-26	12-26	12-26	↗	↗	↗	↗	
	IGBTs	→	→	→	→	50	47-52	14-20	14-24	14-24	14-24	14-24	↗	↗	↗	↗	
	8-bit MCU	↗	↗	↗	↗	30	Allocation	15-33	30	30	40-52	40-52	↗	↗	↗	↗	
	32-bit MCU F0	↗	→	→	→	12-14	Allocation	15-28	30	30	40-52	40-52	↗	↗	↗	↗	
	32-bit MCU F1	↗	→	→	→	12-14	Allocation	15-28	30	30	40-52	40-52	↗	↗	↗	↗	
	32-bit MCU L	↗	→	→	→	12-14	Allocation	15-28	30	30	40-52	40-52	↗	↗	↗	↗	
	Automotive high-end	→	→	→	→	50	Allocation	40-52	40-52	40-52	40-52	40-52	→	→	→	→	
	32-bit MPU	→	→	↗	↗	NA	40	13-16	16-20	16-20	30-52	30-52	↗	↗	↗	↗	

Source : Future Electronics, Deutsche Bank; as of August 14th 2026



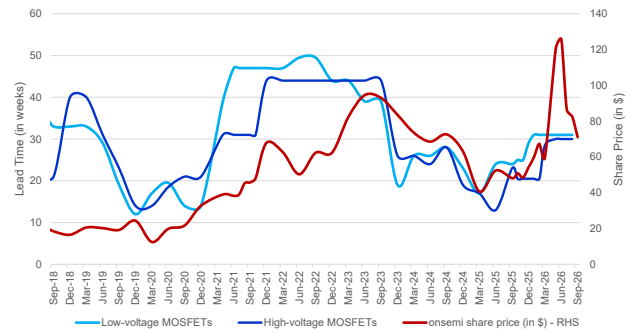
Share performance vs. lead times of key products

Figure 6: Infineon - Lead time vs share price



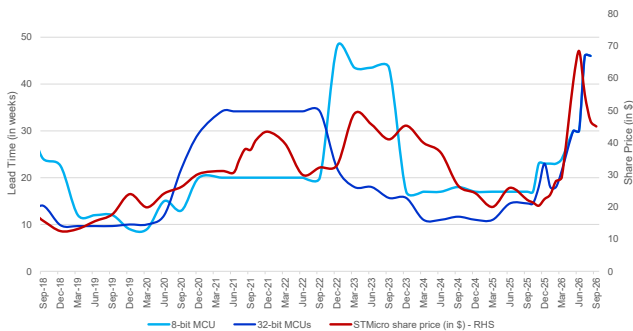
Source : Future Electronics, Bloomberg Finance LP, Deutsche Bank

Figure 7: onsemi - Lead time vs. share price



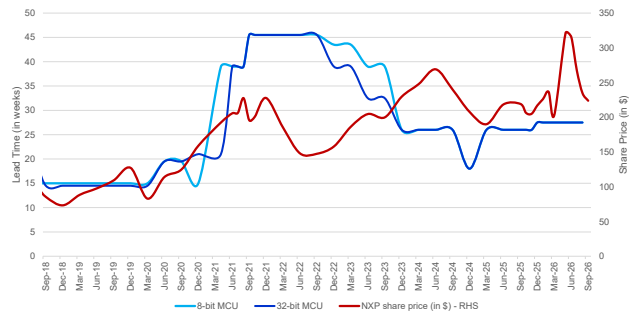
Source :Future Electronics, Bloomberg Finance LP, Deutsche Bank

Figure 8: ST Micro - Lead time vs. share price



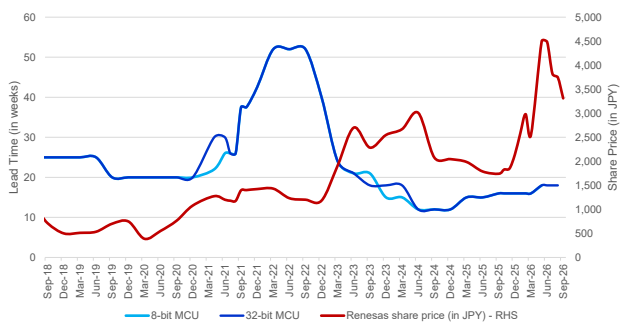
Source :Future Electronics, Bloomberg Finance LP, Deutsche Bank

Figure 9: NXP - Lead time vs. share price



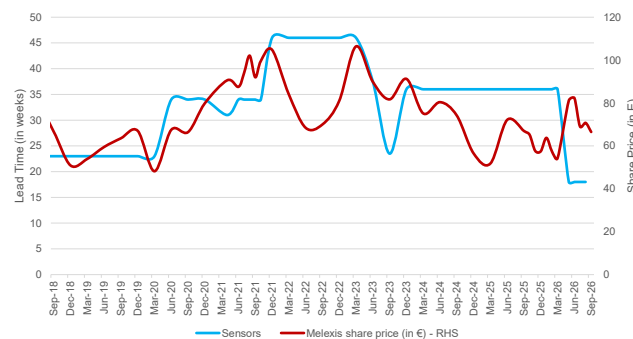
Source :Future Electronics, Bloomberg Finance LP, Deutsche Bank

Figure 10: Renesas - Lead time vs. share price



Source :Future Electronics, Bloomberg Finance LP, Deutsche Bank

Figure 11: Melexis - Lead time vs. share price



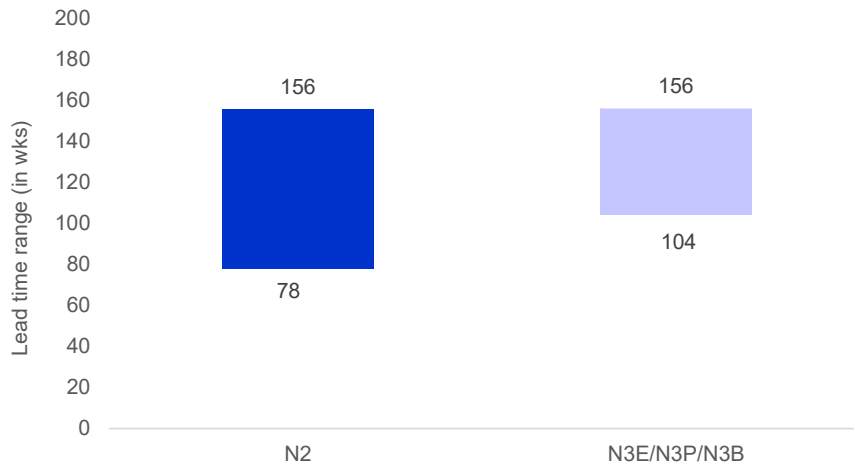
Source :Future Electronics, Bloomberg Finance LP, Deutsche Bank



2. Leading-edge tightness

a) Foundry tightness

Figure 12: N2 & N3 lead time range (in weeks)



Source : Silicon Analysts - [link](#), Deutsche Bank research

Figure 13: IBS view of the shortage: percent of supply versus demand

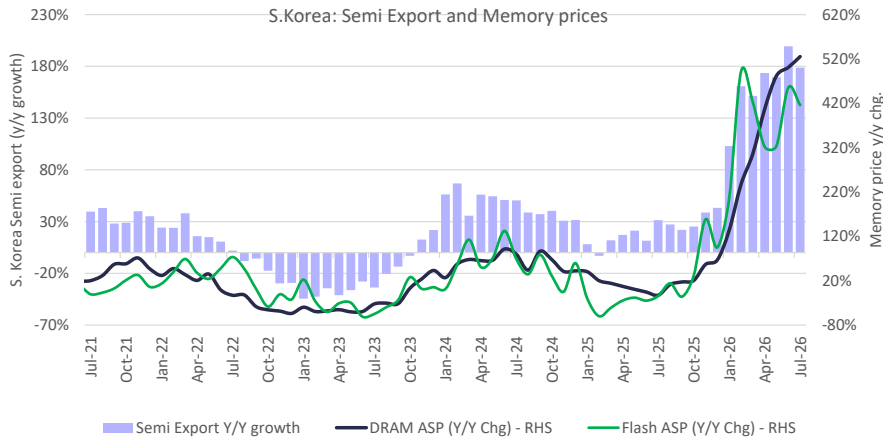
Logic		2026				2027			
		Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
<=2nm	Accelerator	89	88	86	80	75	73	72	75
	CPU	76	76	74	71	70	70	71	73
	Smartphone chipset	90	92	85	79	73	73	72	74
5/4/3nm	Accelerator	74	73	72	73	75	77	79	81
	CPU	75	74	75	76	77	78	80	82
	Miscellaneous	73	72	71	72	73	75	77	79
7nm	Miscellaneous	82	80	78	79	81	82	83	84
16/14/12nm	MPU	105	104	103	102	102	102	103	104
	Miscellaneous	103	103	102	102	101	102	103	104
28/22nm	MCU	107	108	108	107	107	108	109	109
	Miscellaneous	108	108	108	107	107	108	109	109
>28nm	Precision analog	106	102	100	99	98	98	97	96
	PMIC	101	102	101	100	100	99	98	98
	Power transistor	106	103	101	100	99	99	99	98
	Image sensor	110	109	108	106	107	108	107	105
	Miscellaneous	106	104	104	104	104	105	106	107
DRAM									
NA	Discrete products	60	60	60	62	63	64	66	68
NA	HBM	58	57	57	58	60	62	65	67
NAND									
NA	Discrete products	84	85	84	84	85	86	86	87
NA	Solid state drive	81	80	79	78	77	78	79	81

Source : IBS, Deutsche Bank research



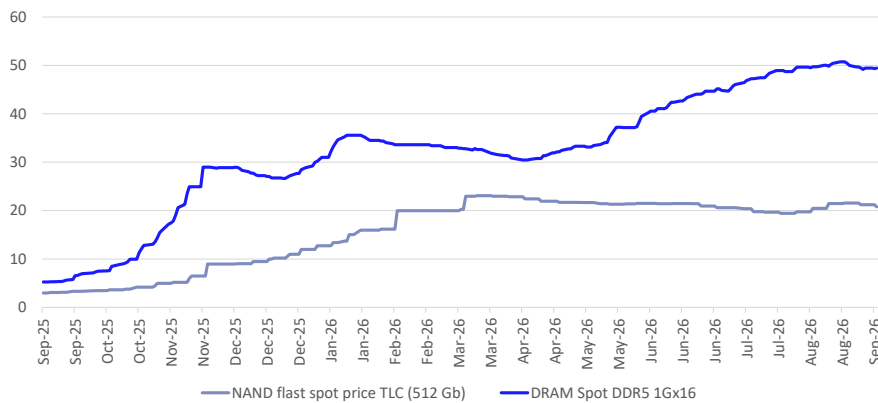
b) Memory pricing trends

Figure 14: Semi export and memory prices, South Korea



Source : Deutsche Bank, South Korean trade statistics, as of July 2026

Figure 15: Spot prices of DDR5 DRAM and NAND flash, last 12M (\$)



Source : Inspectrum, September 9th 2026, Deutsche Bank research



c) Memory supply/demand

Figure 16: DRAM supply/demand model

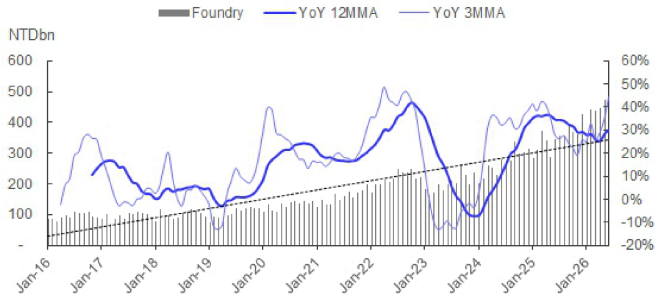
Forecasting Required DRAM WSPM Needed to Support DRAM Bit Growth									
	2023	2024	2025	Forecast					25 - 30 Growth
				2026	2027	2028	2029	2030	
Standard DRAM WSPM Requirements (k WSPM)	1,356	1,490	1,571	1,743	2,093	2,355	2,409	2,458	+887k
y/y Standard DRAM Bit Demand Growth (%)		+10%	+5%	+24%	+33%	+23%	+13%	+13%	
(+) Uplift due to Higher HBM Trade Ratio						N/A			
(-) Impact from Bit / Wafer Efficiencies				-8%	-8%	-6%	-6%	-6%	
(-) Impact from Yield Improvements				-5%	-5%	-5%	-5%	-5%	
= Assumed y/y Standard DRAM WSPM Growth (%)				+11%	+20%	+12%	+2%	+2%	
as a % of Total DRAM WSPM	95%	85%	82%	77%	73%	66%	65%	65%	
HBM WSPM Requirements (k WSPM)	70	270	350	518	774	1,208	1,317	1,317	+967k
y/y HBM Bit Demand Growth (%)		+286%	+30%	+56%	+55%	+67%	+20%	+11%	
(+) Uplift due to Higher HBM Trade Ratio				+5%	+8%	+3%	+3%	+3%	
(-) Impact from Bit / Wafer Efficiencies				-8%	-8%	-6%	-6%	-6%	
(-) Impact from Yield Improvements				-5%	-5%	-8%	-8%	-8%	
= Assumed y/y HBM WSPM Growth (%)				+48%	+50%	+56%	+9%	+0%	
as a % of Total DRAM WSPM	5%	15%	18%	23%	27%	34%	35%	35%	
Total DRAM WSPM Requirements (k WSPM)	1,426	1,760	1,921	2,261	2,868	3,563	3,726	3,775	+14%
Estimated DRAM WSPM Capacity (k WSPM)			1,921	2,051	2,361	2,769	3,151	3,396	+12%
Surplus / (Shortfall)				-210k	-507k	-795k	-575k	-379k	
Demand / Supply (%)				110%	121%	129%	118%	111%	

Source : Deutsche Bank research, Gartner, as of June 2026 - [link](#)



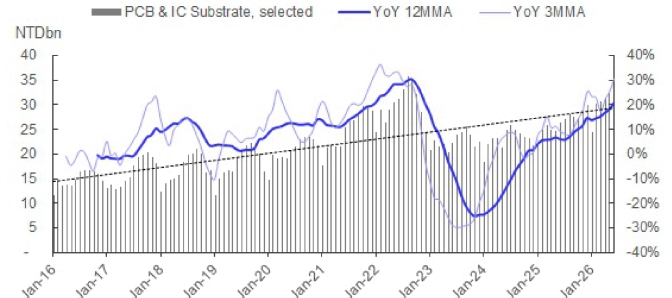
d) Taiwan sales trends

Figure 17: Foundry sales trends



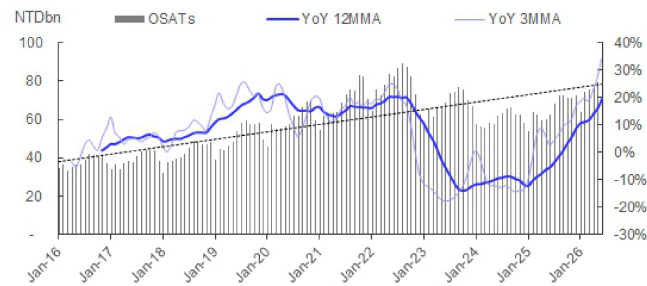
Source : Company data, Deutsche Bank research

Figure 18: PCB/IC substrate sales trends



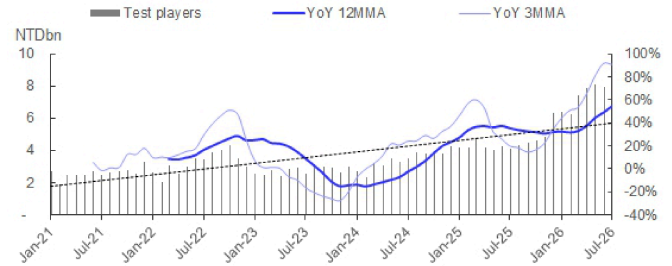
Source : Company data, Deutsche Bank research

Figure 19: OSAT trends



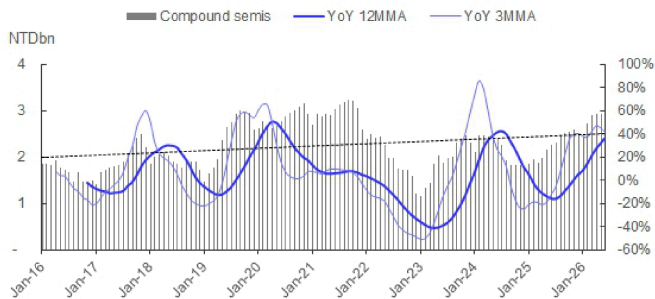
Source : Company data, Deutsche Bank research

Figure 20: Test player trends



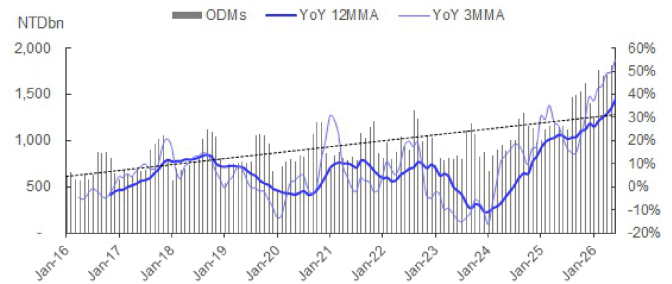
Source : Company data, Deutsche Bank research

Figure 21: Compound semis trends



Source : Company data, Deutsche Bank research

Figure 22: ODM sales trends



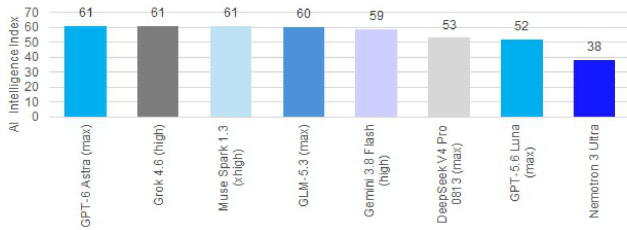
Source : Company data, Deutsche Bank research



3. AI trade monitor

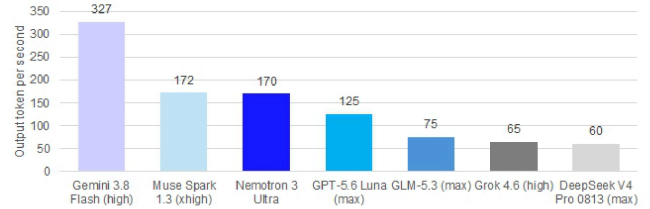
1/ Frontier model performance and monetization

Figure 23: Selected frontier models - Intelligence



Source : Artificial Analysis, Deutsche Bank research, NB list not exhaustive - [link](#)

Figure 24: Selected frontier models - Speed



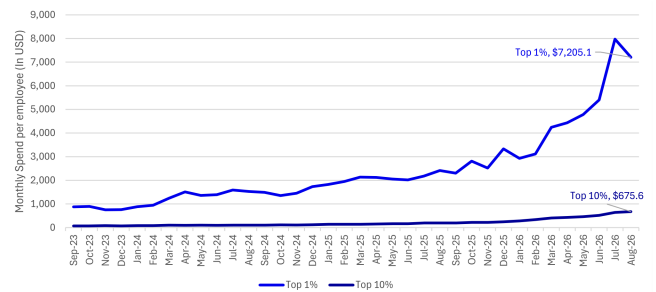
Source : Artificial Analysis, Deutsche Bank research, NB list not exhaustive

Figure 25: Selected frontier models - cost per task (\$)



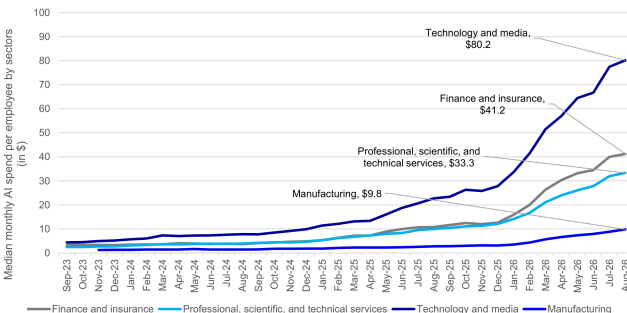
Source : Artificial Analysis, Deutsche Bank research, NB list not exhaustive

Figure 26: RampAI Index, USA: Spend per employee



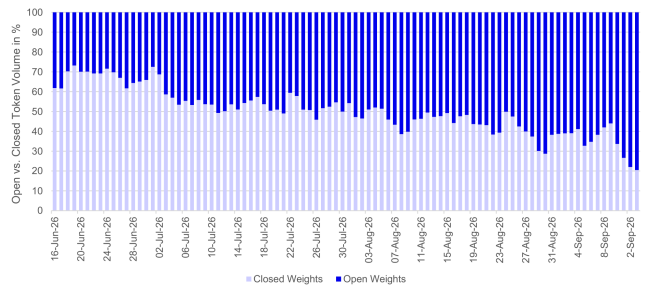
Source : Ramp, , Deutsche Bank research, note for US only - [link](#)

Figure 27: RampAI Index: Spend per employee by sector



Source : Ramp, , Deutsche Bank research, note for US only

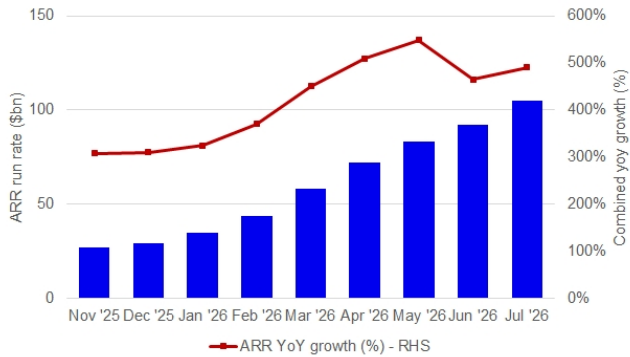
Figure 28: Open weights vs closed weight token adoption trends





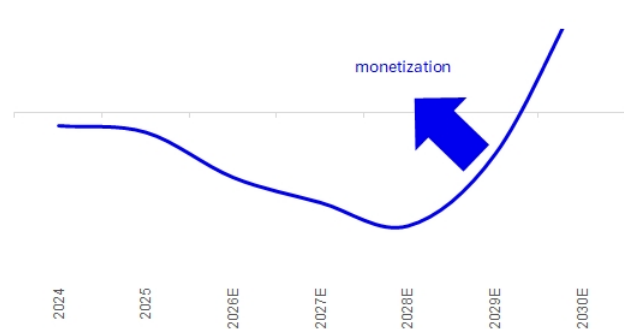
2/ Financial risks to the AI trade

Figure 29: Top frontier models - ARR growth



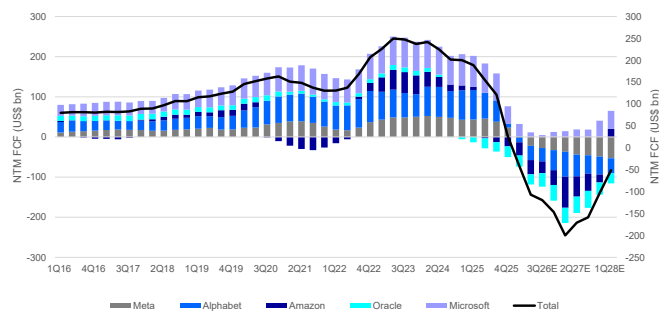
Source : Indicative trends as per press reports - eg Bloomberg Finance LP, Reuters, CNBC

Figure 30: Top frontier models - FCF, indicative



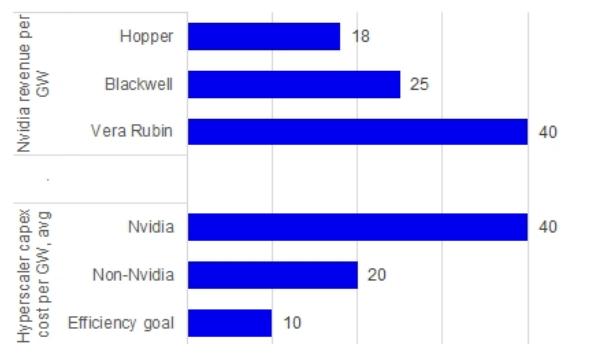
Source : Indicative as per press reports including WSJ

Figure 31: Major hyperscaler FCF



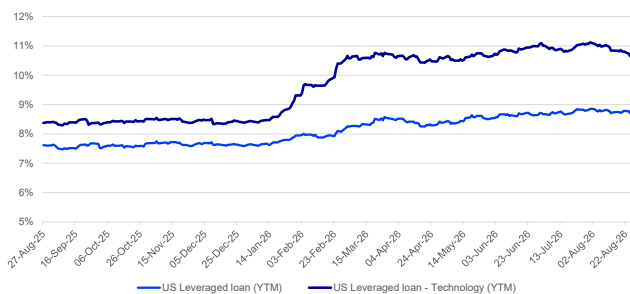
Source : Company data, Bloomberg Finance LP

Figure 32: Revenue, capex per GW (\$bn)



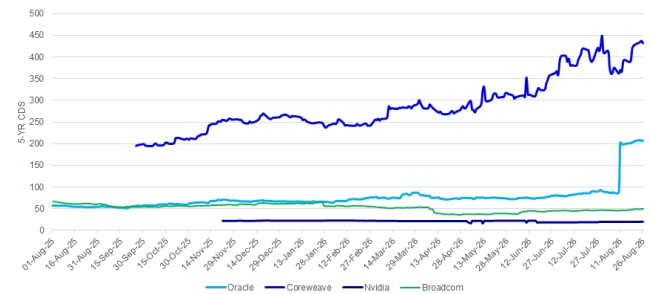
Source : IBS, company data

Figure 33: US leveraged loan YTM - Tech vs all



Source : Deutsche Bank research, Bloomberg Finance LP - for DB's views on private credit - see [here](#)

Figure 34: Financing/default risk, CDS - major players

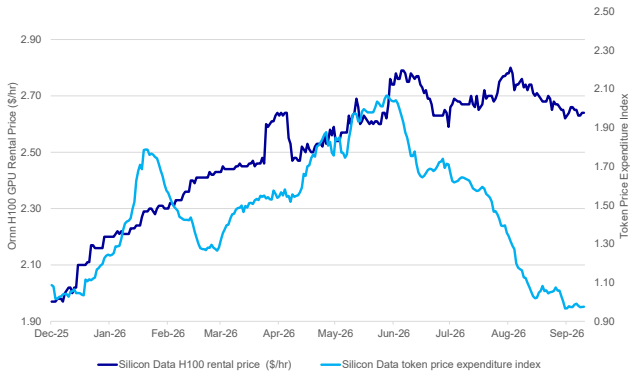


Source : Deutsche Bank research, Bloomberg Finance LP



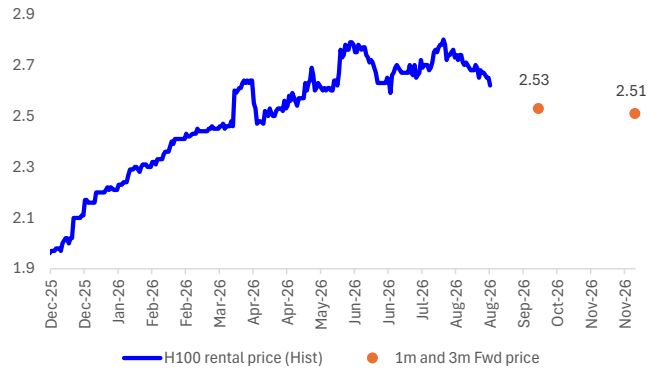
3/ Data center GPU rental and token pricing

Figure 35: GPU rental vs token pricing (\$ / m tokens)



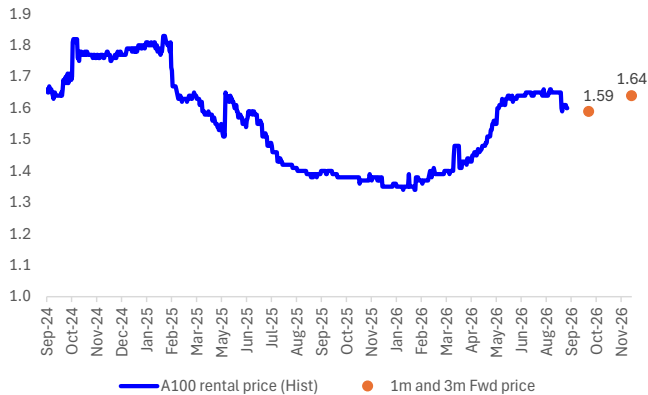
Source : Deutsche Bank research, Bloomberg Finance LP

Figure 36: Nvidia H100 rental price (\$ per hour)



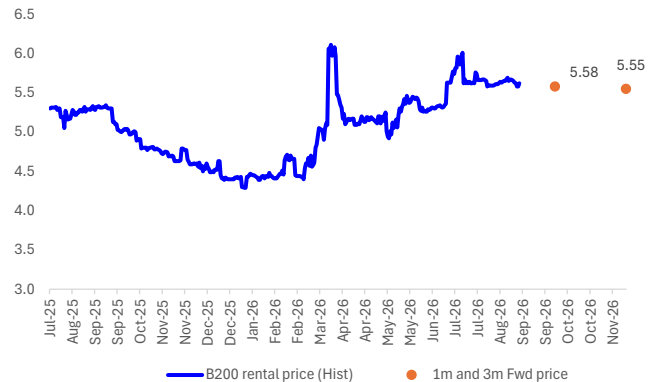
Source : Deutsche Bank, Bloomberg Finance LP, Silicon Data

Figure 37: Nvidia A100 rental price (\$ per hour)



Source : Deutsche Bank, Bloomberg Finance LP, Silicon Data

Figure 38: Nvidia B200 rental price (\$ per hour)

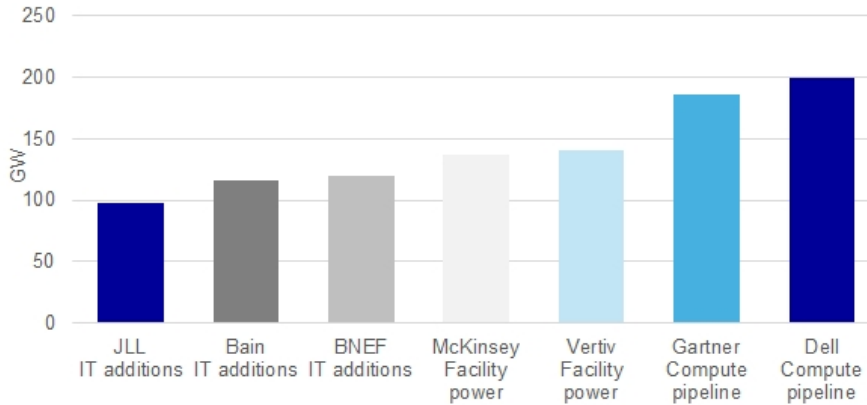


Source : Deutsche Bank, Bloomberg Finance LP, Silicon Data



4. Industrial constraints

Figure 39: Data center GW builds, 2026-30E cumulative, 5 year period



Source : JLL, Bain, BNEF, McKinsey, Vertiv, Gartner, Dell

Figure 40: Industrial pinchpoints for data center spend

Industrial equipment	2026 lead time	Historical baseline
Medium voltage switchgear, 15kV	52 to 80 weeks	About 24 weeks
Medium voltage switchgear, 38kV	78 to 104 weeks	About 24 weeks
Pad mount transformers, 0 to 5 MVA	40 to 65 weeks	About 8 weeks
Substation transformers, 5 to 50 MVA	75 to 110 weeks	Not stated
Generator step-up, above 50 MVA	100 to 150+ weeks	Not stated
Diesel generators, above 3,000 kW	90 to 110 weeks	About 20 weeks
UPS, 500 to 1,500 kVA	30 to 48 weeks	Not stated

Source : Industrial Sage, Terrapin Construction Group - [link](#)

Figure 41: The gas turbine major bottleneck

Gas turbines	Metric
Gas turbine backlog, Q2 2026	116 GW
Gas turbine backlog, Q1 2026	100 GW
Current annual output	Roughly 20 GW
Planned annual output by 2030	30 GW
Expected backlog plus reservations, end	At least 125 GW
Reservations now open for	2031 delivery
Data center share of backlog	About 20%

Source : GE Vernova



Appendix 1

Important Disclosures

*Other information available upon request

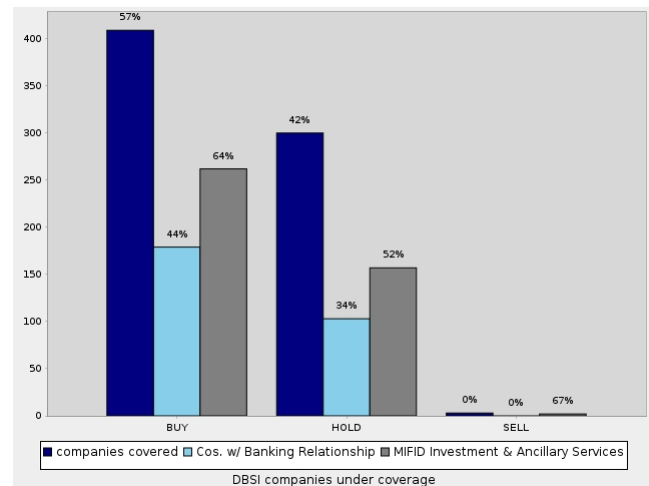
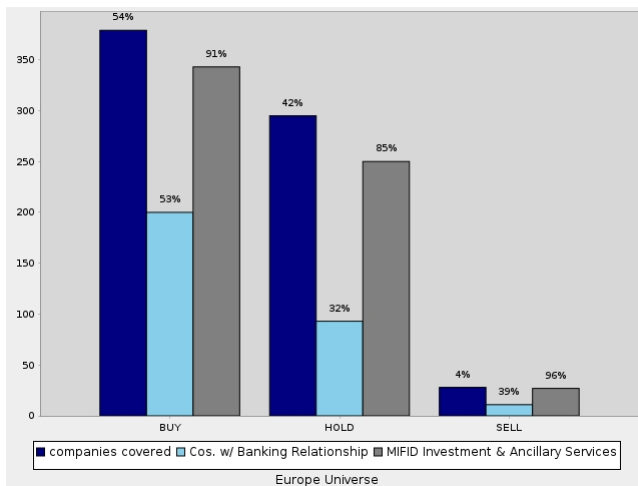
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Equity rating dispersion and banking relationships



Equity Rating and Dispersion Key

The Equity Rating Dispersion Chart depicts the following:

The proportion of recommendations that are rated "buy", "sell" and "hold" over the previous 12 months. This is shown for securities issued in the stated region e.g. "Europe Universe". See rating definitions below. This is represented by the "Companies Covered" bars in the chart. The percentage value displayed above the bar is the proportion as a percentage. E.g. 50% above the "buy" / "Companies Covered" bar means that 50% of DB's equity research covered companies over the past 12 months have a "buy" rating.

Next to each of the three respective bars showing the proportion of "buy", "sell" and "hold" recommendations we provide two additional bars to show:

- The proportion of "buy", "sell" or "hold" recommendations where Deutsche Bank and/or Affiliates provided MIFID Investment or Ancillary Services in the past 12 months. This is represented in the "MIFID Investment and Ancillary Services" bar. The percentage value displayed above the bar shows the proportion of Companies Covered with the given rating where DB has also provided MIFID Investment and Ancillary Services in the past 12 months. E.g. 50% above the "Cos. w/ MIFID Investment and Ancillary Services" bar means 50% of the Companies Covered with the rating stated have also received MIFID Investment and Ancillary Services from DB.

- The proportion of "buy" (or "sell" or "hold") recommendations where Deutsche Bank and/or Affiliates has provided Investment Banking services in the past 12 months for which it has received compensation. The percentage value displayed above the bar shows the proportion of Companies Covered with the stated rating where DB has also provided Investment Banking services in the past 12 months. E.g. 50% above the "Cos. w/ Investment Banking relationship" bar means 50% of the Companies Covered with the rating stated also have an Investment Banking Relationship with DB.

Buy: Based on a current 12-month view of TSR, we recommend that investors buy the stock.

Sell: Based on a current 12-month view of TSR, we recommend that investors sell the stock.

Hold: We take a neutral view on the stock 12-months out and, based on this time horizon, do not recommend either a Buy or Sell.

TSR = Total Shareholder Return. Percentage change in share price from current price to projected target price plus projected dividend yield

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